

Chapter 4 SUBNETS

Final - গ্রুপ জিলাবাস

PDF

① IP Address

② Interface

③ Subnet

→ Router has multiples interfaces
Host has only one

① NAT

② DHCP

③ CIDR

④ IP Address structure

→ Subnet Part → common high order bits

→ Host Part → low order bits

How To Subnet?

① Delete Router & Hosts

→ after that each island of
end devices are called 'subnet'

NAT → Network Address Translation

Subnet Mask : /24

High-order 24 bits : Subnet part of IP Address

① Where are the subnets?

② What are the /24 subnet addresses?

3 routers → 3 subnets

But in example,

3 routers → 6 subnets

Exam Task:

Find subnets from a complex network

→ also find out IP addresses

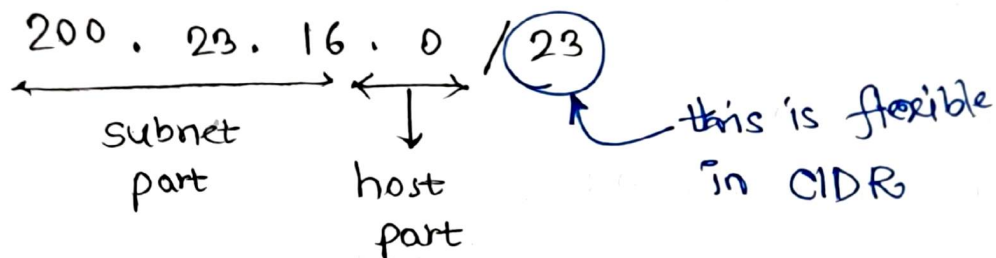
⑤ CIDR

CIDR: Classless InterDomain Routing << cidr >>

↳ subnet portion of address of arbitrary length

↳ address format:

$a.b.c.d/x$, $x = \# \text{ of bits in subnet part}$



IP Class

A → 8 bit

B → 16 bit

C → 24 bit

⑥ IP Addresses : How to get one?

A. How does a host get IP address within its network? (host part of address)

B. How does a network get IP address for itself? (network part of address)

Ans.:

① hard-coded by sys admin << Manual >>

② DHCP : dynamically get one from server

⑥ DHCP

Dynamic Host Configuration Protocol

↙ also inside a Router

Goal:

host dynamically obtain... << [A] PDF >>

⑦ DHCP: Client-Server Scenario

DHCP server will be collocated in the router

<< ~~see~~ >>

<< scenario from [A] PDF >>

* DHCP server

* DHCP discovery

* DHCP offer

* DHCP request

* DHCP ACK

→ Four step Process

⑧ DHCP: Why it is better than manual?

It can return:

① Address of first hop router for client

② Name & IP of DNS

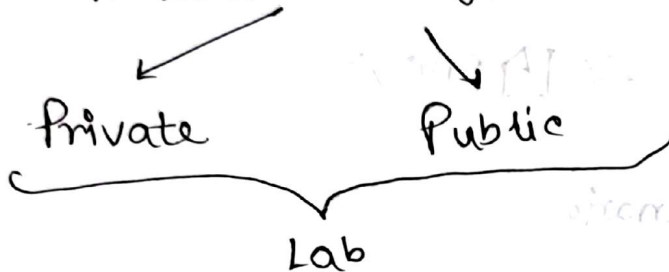
③ Network Mask

⑨ NAT

Shayan

all devices in local network share just one IPv4 address
as far as outside world is concerned

IP address classification



192.168.X.X → Private

→ Public (needs to be purchased)

Private IP --mapping--> Public IP